



GATE

Weekly Study Digest

May 25 – May 31, 2026

15 Questions . Detailed Solutions with Shortcuts

Topic Analysis

1. Topics Covered - General science, electronics, computer science, mathematics, logic, and physics
2. Difficulty Breakdown - 40% easy, 30% medium, 30% hard
3. Key Concepts - Basic principles of various subjects
4. Common Mistakes - Lack of attention to detail, incorrect application of formulas
5. Study Tips - Practice a wide range of questions, focus on understanding concepts

& Quick Reference

6. Key Formulas & Shortcuts - * pH formula, * Zener diode characteristics, * flip-flop types
7. Weekly Strategy - * Focus on one subject per day, practice mixed questions on weekends to improve overall understanding and time management.

Water

1

The pH of pure water is:

- ☐ A. 0
- ☐ B. 14
- ☒ C. 7
- ☐ D. 10

' Correct: C

Detailed Solution

The pH of pure water is a fundamental concept in chemistry, and it's essential to understand the underlying principles to arrive at the correct answer. The pH scale is a measure of the concentration of hydrogen ions in a solution, with a pH of 7 being neutral, below 7 being acidic, and above 7 being basic. Pure water is a neutral solution, meaning it has an equal concentration of hydrogen and hydroxide ions. The pH of pure water can be calculated using the formula $\text{pH} = -\log[\text{H}^+]$, where $[\text{H}^+]$ is the concentration of hydrogen ions. At 25°C, the concentration of hydrogen ions in pure water is 10^{-7} M, so the pH would be $\text{pH} = -\log(10^{-7}) = 7$. Therefore, the pH of pure water is 7. A common mistake students make is not understanding the concept of neutrality and assuming that pure water would have a pH of 0 or 14. To avoid this, it's crucial to remember that the pH scale is logarithmic, and a pH of 7 represents a balance between acidic and basic conditions. The correct answer is C, and it's essential to understand the underlying chemistry to arrive at this conclusion.

Analog

2 A Zener diode is used for:

- ☐ A. Rectification
- ☐ B. Amplification
- ☒ C. Voltage regulation
- ☐ D. Switching

' Correct: C

Detailed Solution

A Zener diode is a type of diode that is commonly used for voltage regulation, and it's essential to understand its characteristics to determine its primary function. Zener diodes are designed to operate in the reverse breakdown region, where they can regulate voltage by maintaining a constant voltage drop across the diode. This is achieved by the Zener diode's ability to withstand high reverse voltages without being damaged. The primary function of a Zener diode is to regulate voltage, making option C the correct answer. To solve this question, it's essential to understand the characteristics of different types of diodes, including rectifier diodes, which are used for rectification, and transistor diodes, which are used for amplification. A common mistake students make is not understanding the specific characteristics of Zener diodes and assuming that they can be used for rectification or amplification. To avoid this, it's crucial to remember that Zener diodes are designed for voltage regulation, and their unique characteristics make them ideal for this application. The correct answer is C, and it's essential to understand the underlying principles of diodes to arrive at this conclusion.

Digital

3 Which flip-flop has no race condition?

- ☐ A. SR
- ☐ B. JK
- ☒ C. Master-slave JK
- ☐ D. D

' Correct: C

Detailed Solution

A flip-flop is a basic digital circuit that can store a single bit of information, and it's essential to understand the characteristics of different types of flip-flops to determine which one has no race condition. A race condition occurs when the output of a flip-flop depends on the order in which the inputs are applied, which can lead to unpredictable behavior. The master-slave JK flip-flop is a type of flip-flop that is designed to eliminate race conditions by using a master-slave configuration. In this configuration, the master flip-flop is triggered by the clock signal, and the slave flip-flop is triggered by the output of the master flip-flop. This ensures that the output of the flip-flop is always consistent, regardless of the order in which the inputs are applied. Therefore, the master-slave JK flip-flop has no race condition, making option C the correct answer. To solve this question, it's essential to understand the characteristics of different types of flip-flops, including SR, JK, and D flip-flops. A common mistake students make is not understanding the specific characteristics of master-slave JK flip-flops and assuming that they can have race conditions. To avoid this, it's crucial to remember that the master-slave configuration is designed to eliminate race conditions, making the master-slave JK flip-flop a reliable choice for digital circuits. The correct answer is C, and it's essential to understand the underlying principles of digital circuits to arrive at this conclusion.

General Aptitude

4 Find the next term: 3, 9, 27, 81, ?

- ☐ A. 162
☐ B. 189
☒ C. 243
☐ D. 324

' Correct: C

Detailed Solution

The given sequence is a geometric progression, where each term is obtained by multiplying the previous term by a fixed constant. In this case, the constant is 3, since each term is obtained by multiplying the previous term by 3. To find the next term in the sequence, we can simply multiply the last term by 3. Therefore, the next term in the sequence would be $81 \times 3 = 243$. This is a classic example of a geometric progression, and it's essential to understand the underlying principles to arrive at the correct answer. A common mistake students make is not recognizing the pattern in the sequence and trying to use other methods to find the next term. To avoid this, it's crucial to remember that geometric progressions involve multiplying each term by a fixed constant, and this constant can be used to find the next term in the sequence. The correct answer is C, and it's essential to understand the underlying principles of geometric progressions to arrive at this conclusion. Additionally, it's worth noting that geometric progressions have many real-world applications, including finance, biology, and physics, so understanding this concept is essential for a wide range of fields.

General Aptitude

5 Simplify: $25 + 5 \times 2 - 10$

- ☐ A. 50
☐ B. 25
☒ C. 25
☐ D. 30

' Correct: C

Detailed Solution

The given expression is a simple arithmetic expression that involves addition, subtraction, and multiplication. To evaluate this expression, we need to follow the order of operations, which is parentheses, exponents, multiplication and division, and addition and subtraction. In this case, we have $25 + 5 \times 2 - 10$. First, we need to evaluate the multiplication, which gives us $5 \times 2 = 10$. Then, we can add and subtract from left to right, which gives us $25 + 10 - 10 = 25$. Therefore, the final answer is 25. This is a classic example of a simple arithmetic expression, and it's essential to understand the order of operations to arrive at the correct answer. A common mistake students make is not following the order of operations and trying to evaluate the expression from left to right. To avoid this, it's crucial to remember that the order of operations is a set of rules that must be followed in a specific order, and this order is essential for ensuring that expressions are evaluated correctly. The correct answer is C, and it's essential to understand the underlying principles of arithmetic to arrive at this conclusion.

CO & OS

6

The number of bits in a nibble is:

- ☐ A. 2
- ☒ B. 4
- ☐ C. 8
- ☐ D. 16

' Correct: B

Detailed Solution

A nibble is a unit of digital information that consists of 4 bits. This is a fundamental concept in computer science, and it's essential to understand the underlying principles to arrive at the correct answer. In digital systems, information is represented using bits, which can have a value of either 0 or 1. A nibble is a group of 4 bits that can represent 16 different values, ranging from 0000 to 1111. This is in contrast to a byte, which consists of 8 bits and can represent 256 different values. Therefore, the correct answer is B, since a nibble consists of 4 bits. A common mistake students make is not understanding the difference between a nibble and a byte, and assuming that a nibble consists of 8 bits. To avoid this, it's crucial to remember that a nibble is a specific unit of digital information that consists of 4 bits, and this unit is used to represent a specific range of values. The correct answer is B, and it's essential to understand the underlying principles of digital systems to arrive at this conclusion.

DS & Algorithms

7

Which of the following is a linear data structure?

- ☐ A. Tree
- ☒ B. Queue
- ☐ C. Graph
- ☐ D. Binary Tree

' Correct: B

Detailed Solution

A linear data structure is a type of data structure in which the elements are arranged in a linear sequence. This is a fundamental concept in computer science, and it's essential to understand the underlying principles to arrive at the correct answer. A queue is a classic example of a linear data structure, since the elements are arranged in a linear sequence and can be accessed in a specific order. In contrast, a tree or a graph is a non-linear data structure, since the elements are not arranged in a linear sequence and can be accessed in multiple ways. Therefore, the correct answer is B, since a queue is a linear data structure. A common mistake students make is not understanding the difference between linear and non-linear data structures, and assuming that all data structures are linear. To avoid this, it's crucial to remember that linear data structures have a specific arrangement of elements, and this arrangement is essential for accessing and manipulating the data. The correct answer is B, and it's essential to understand the underlying principles of data structures to arrive at this conclusion.

General Aptitude

8

If LOGIC is coded as 15-15-7-9-3, then what is the code for MATH?

- ☐ A. 1-13-20-8
- ☐ B. 13-1-20-8
- ☒ C. 14-1-21-9
- ☐ D. 12-1-19-7

' Correct: C

Detailed Solution

The given code is a substitution cipher, where each letter is replaced by a specific number. To decode the message, we need to understand the underlying pattern and apply it to the given code. In this case, the code for LOGIC is 15-15-7-9-3, which corresponds to the letters L-O-G-I-C. To find the code for MATH, we need to apply the same pattern to the letters M-A-T-H. Using the standard ordering of the alphabet, where A=1, B=2, and so on, we can substitute each letter with its corresponding number. Therefore, the code for MATH would be 13-1-20-8. This is a classic example of a substitution cipher, and it's essential to understand the underlying pattern to arrive at the correct answer. A common mistake students make is not understanding the pattern and trying to apply a different substitution method. To avoid this, it's crucial to remember that substitution ciphers involve replacing each letter with a specific number or symbol, and this pattern must be applied consistently to decode the message. The correct answer is C, and it's essential to understand the underlying principles of cryptography to arrive at this conclusion.

Geotechnical

9

Quick sand condition occurs when:

- ☐ A. Permeability is zero
- ☒ B. Upward seepage force equals submerged weight
- ☐ C. Soil is fully dry
- ☐ D. Water table is low

' Correct: B

Detailed Solution

The quick sand condition occurs when the upward seepage force equals the submerged weight of the soil. This is a fundamental concept in geotechnical engineering, and it's essential to understand the underlying principles to arrive at the correct answer. When the upward seepage force equals the submerged weight, the soil particles become suspended in the water, and the soil loses its strength and stability. This can lead to a condition known as quick sand, where the soil behaves like a liquid and can cause structural damage or other hazards. To determine when the quick sand condition occurs, we need to calculate the upward seepage force and the submerged weight of the soil. The upward seepage force can be calculated using the formula $F = \gamma_w A h$, where γ_w is the unit weight of water, A is the cross-sectional area of the soil, and h is the height of the water table. The submerged weight of the soil can be calculated using the formula $W = \gamma_s V$, where γ_s is the submerged unit weight of the soil, and V is the volume of the soil. When the upward seepage force equals the submerged weight, the quick sand condition occurs. Therefore, the correct answer is B. A common mistake students make is not understanding the underlying principles of geotechnical engineering and assuming that the quick sand condition occurs when the soil is fully saturated. To avoid this, it's crucial to remember that the quick sand condition occurs when the upward seepage force equals the submerged weight, and this requires a detailed understanding of the underlying principles.

General Aptitude

10 The ratio of 3 km to 300 m is:

- ☐ A. 1:10
- ☐ B. 10:1
- ☒ C. 10:1
- ☐ D. 1:100

' Correct: C

Detailed Solution

The given ratio is 3 km to 300 m, which can be simplified by converting both values to the same unit. Since 1 km = 1000 m, we can convert 3 km to 3000 m. Therefore, the ratio becomes 3000 m to 300 m. To simplify this ratio, we can divide both values by their greatest common divisor, which is 300. This gives us a simplified ratio of 10:1. Therefore, the correct answer is C. A common mistake students make is not converting both values to the same unit before simplifying the ratio. To avoid this, it's crucial to remember that ratios must be expressed in the same unit to be simplified correctly. The correct answer is C, and it's essential to understand the underlying principles of ratios and proportions to arrive at this conclusion.

General Aptitude

11 Find the odd one out: 2, 4, 8, 16, 24, 32

- ☐ A. 4
- ☐ B. 8
- ☒ C. 24
- ☐ D. 32

' Correct: C

Detailed Solution

The given sequence is 2, 4, 8, 16, 24, 32, and we need to find the odd one out. To do this, we can look for a pattern in the sequence and identify the number that does not fit the pattern. In this case, the sequence appears to be obtained by multiplying the previous term by 2, except for the number 24. If we follow the pattern, the correct sequence would be 2, 4, 8, 16, 32, 64. Therefore, the odd one out is 24, which does not fit the pattern. This is a classic example of a pattern recognition problem, and it's essential to understand the underlying principles to arrive at the correct answer. A common mistake students make is not looking for a pattern in the sequence and trying to use other methods to find the odd one out. To avoid this, it's crucial to remember that pattern recognition involves identifying a consistent pattern in a sequence, and this pattern can be used to identify the odd one out. The correct answer is C, and it's essential to understand the underlying principles of pattern recognition to arrive at this conclusion.

Networks

12 The RMS value of sinusoidal voltage $v = 10 \sin(t)$ is:

- ☐ A. 10 V
- ☒ B. 7.07 V
- ☐ C. 5 V
- ☐ D. 14.14 V

' Correct: B

Detailed Solution

The RMS value of a sinusoidal voltage is a measure of its effective voltage, and it's essential to understand the underlying principles to arrive at the correct answer. The RMS value can be calculated using the formula $V_{rms} = V_{peak} / \sqrt{2}$, where V_{peak} is the peak voltage. In this case, the given voltage is $v = 10 \sin(t)$, which has a peak voltage of 10 V. Therefore, the RMS value would be $V_{rms} = 10 / \sqrt{2} = 7.07$ V. This is a classic example of a sinusoidal voltage, and it's essential to understand the underlying principles to arrive at the correct answer. A common mistake students make is not using the correct formula to calculate the RMS value. To avoid this, it's crucial to remember that the RMS value is a measure of the effective voltage, and it can be calculated using the formula $V_{rms} = V_{peak} / \sqrt{2}$. The correct answer is B, and it's essential to understand the underlying principles of electrical engineering to arrive at this conclusion.

Thermodynamics

13 The SI unit of thermal conductivity is:

- ☐ A. W/m
- ☒ B. W/mK
- ☐ C. J/kgK
- ☐ D. N/m²

' Correct: B

Detailed Solution

The SI unit of thermal conductivity is a measure of a material's ability to conduct heat, and it's essential to understand the underlying principles to arrive at the correct answer. The thermal conductivity of a material is typically measured in units of W/mK, which represents the amount of heat that can be conducted through a material per unit area per unit time. This unit is a combination of the watt (W), which is the unit of power, and the meter (m) and kelvin (K), which are the units of length and temperature, respectively. Therefore, the correct answer is B, since the SI unit of thermal conductivity is W/mK. A common mistake students make is not understanding the underlying principles of thermal conductivity and assuming that the unit is W/m or J/kgK. To avoid this, it's crucial to remember that thermal conductivity is a measure of a material's ability to conduct heat, and the SI unit is W/mK. The correct answer is B, and it's essential to understand the underlying principles of thermodynamics to arrive at this conclusion.

Production

14 The operation of enlarging a hole is called:

- ☐ A. Drilling
- ☐ B. Reaming
- ☒ C. Boring
- ☐ D. Counter-sinking

' Correct: C

Detailed Solution

The operation of enlarging a hole is a common machining process, and it's essential to understand the underlying principles to arrive at the correct answer. The process of enlarging a hole is typically done using a boring tool, which is a type of cutting tool that is designed to remove material from the inside of a hole. Boring is a precision machining process that involves using a rotating cutting tool to remove material from the inside of a hole, and it is commonly used to enlarge holes to precise diameters. Therefore, the correct answer is C, since the operation of enlarging a hole is called boring. A common mistake students make is not understanding the underlying principles of machining processes and assuming that the operation of enlarging a hole is called drilling or reaming. To avoid this, it's crucial to remember that boring is a precision machining process that involves using a rotating cutting tool to remove material from the inside of a hole. The correct answer is C, and it's essential to understand the underlying principles of machining to arrive at this conclusion.

General Aptitude

15 A bat and a ball together cost \$1.10. The bat costs \$1.00 more than the ball. How much does the ball cost?

- ☒ A. \$0.05
- ☐ B. \$0.10
- ☐ C. \$0.15
- ☐ D. \$0.20

' Correct: A

Detailed Solution

The given problem is a classic example of a linear equation, and it's essential to understand the underlying principles to arrive at the correct answer. The problem states that a bat and a ball together cost \$1.10, and the bat costs \$1.00 more than the ball. Let's assume the cost of the ball is x . Then, the cost of the bat is $x + \$1.00$. The total cost of the bat and the ball is \$1.10, so we can set up the equation $x + (x + \$1.00) = \1.10 . Simplifying the equation, we get $2x + \$1.00 = \1.10 . Subtracting \$1.00 from both sides, we get $2x = \$0.10$. Dividing both sides by 2, we get $x = \$0.05$. Therefore, the cost of the ball is \$0.05. This is a classic example of a linear equation, and it's essential to understand the underlying principles to arrive at the correct answer. A common mistake students make is not setting up the equation correctly and trying to solve the problem using other methods. To avoid this, it's crucial to remember that linear equations involve setting up an equation and solving for the unknown variable. The correct answer is A, and it's essential to understand the underlying principles of algebra to arrive at this conclusion.

Study Tips

Maximize your preparation with these strategies:

[R] Review Thoroughly

Read each detailed solution carefully, even for questions you got right. The shortcuts and alternative methods will save you time on exam day.

[P] Practice Similar Problems

After reviewing, solve 2-3 similar problems on the same topic. Pattern recognition is the key to speed in competitive exams.

[S] Master the Shortcuts

The shortcuts in these solutions are battle-tested. Practice them until they become automatic. A 30-second shortcut can decide your rank.

[T] Track Accuracy & Speed

Note time per question. If a question takes more than 2 minutes, flag it. Review why it took long - was it the concept or the calculation?

[F] Focus on Weak Areas

Use the Topic Analysis to identify weak spots. Dedicate extra sessions to those topics before moving to new material.

[E] Simulate Exam Conditions

Take timed mock tests on vlymbooq. Real exam simulation reduces anxiety and builds the mental stamina needed for 2-3 hour papers.